

The First Perceval Quest - Phase II

Team *Qaradog*: Ludovic Le Frioux, Valentin Macheret, Antoine Radet

March 28th, 2025

Introduction

How can photonic-based quantum computing improve the performance of MNIST classification? This is the question asked by the QML challenge¹ powered by *Quandela* and *Scaleway*. In Phase I we explored various solutions using our modular hybrid ML framework. We used PCA for dimension reduction to go from a 27x27 pixels image to 12 values that can be encoded into a 12-mode photonic circuit, without significant success.

For Phase II, we present the following contributions: (i) integration of a differentiable quantum layer into our framework, (ii) implementation of a parametric quantum convolutional (qconv2d) pipeline within our framework, and (iii) demonstration that the quantum pipeline is able to learn, showing promise for future development.

Technical Description

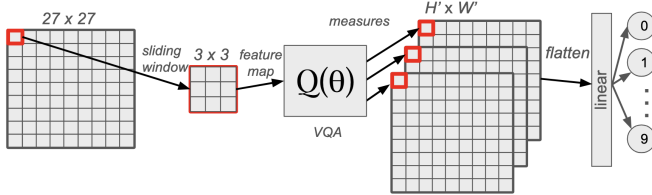


Fig.1: A model architecture including the qconv2d layer

2D convolution² is a common layer in neural networks for vision applications. Inspired by *S. Shi, et al.*³, our implementation (Fig. 1) replaces the dot product of traditional convolution by a quantum circuit while keeping the sliding window principle. Our quantum implementation is a photonic circuit with two parts: a feature map and an ansatz. The *feature map* uses a fixed input Fock state and contains beam splitters (BS) and phase shifters (PS), with some parameters that are fixed and others that depend on input pixel values. The *ansatz* consists of BS and PS whose parameters are learned during training using Adam optimizer. For both parts, two architectures were considered.

For the feature map, the “*Achilles*” architecture, based on our Phase I research, uses a fixed set of BS to dispatch 2 photons over all modes, followed by one PS on each mode whose angle is set by the pixel input values x using: $(x - 0.5) \times \frac{\pi}{2}$. The “*Odysseus*” approach, uses 4 photons and 27 components (BS.H, BS.Ry, and PS) parametrized by input x using 3 variants ($-2\pi x$, $2\pi x$, $\sin(2\pi x)$).

For the ansatz, the “*Penarddun*” architecture consists of a rectangle arrangement of Mach-Zender interferometers with a depth of 6, totaling 48 learnable parameters. The “*Gofanon*” architecture is made up of repeating BS.H + 2PS and BS.Ry + 2PS for a total of 96 variable parameters.

Only Fock states are used as circuit input and output. In simulation, we get the probability of each possible output Fock state. The quantum convolution layer outputs m matrices, one per mode. If we retain the most probable output Fock state, these matrices hold the output photons count for each associated mode, ranging from 0 to the number of photons. Alternatively, we computed the “average photon probability” per mode by multiplying and adding each Fock state’s coordinates by its probability, resulting in matrices with positive floating-point values.

Results and Analysis

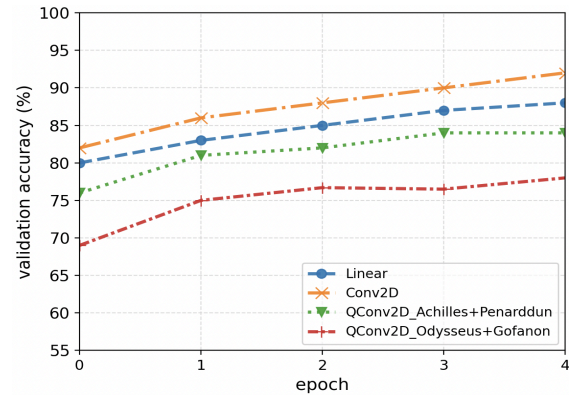


Fig.2: Models validation accuracy over 5 epochs

We simulated the photonic circuit using a differentiable quantum layer instead of Perceval’s usual Processor. The circuit is converted into PyTorch unitary matrices which can then be run and differentiated with usual PyTorch tools, enabling batch computation to reduce training times.

Regardless of the feature map (all variables *vs.* fixed circuit with phase encoding) or output interpretation (most probable Fock state *vs.* cumulative sum), our model successfully learned during training.

Quantum implementations lead to lower validation accuracy, but converges faster than classical models as shown on Fig. 2. (the classic models stop improving after 5 epochs, though it is not reflected on the above figure).

Also, using the average photon probabilities does not bring more value: it is far slower to compute, and results in lower accuracy (50-60% final validation accuracy). However, it could reduce the number of runs needed on a real QPU compared to finding the most probable output which requires running many shots.

Conclusion

In Phase II, we replaced PCA pre-processing with a convolutional approach in order to maintain a small circuit size, small training time (around 10min/epoch) and achieve localization. We also noticed that small feature maps and ansatz are enough to learn the classical MNIST with a unique kernel, instead of multiple kernels in usual 2D convolution.

¹ <https://github.com/Quandela/HybridAIQuantum-Challenge>

² <https://pytorch.org/docs/stable/generated/torch.nn.Conv2d.html>

³ *S. Shi, et al. "Hybrid quantum-classical convolutional neural network for phytoplankton classification." Frontiers in Marine Science 10 (2023)*